

Global Weather Repository

Analysis & Forecasting

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1. Introduction

Weather shapes almost every consequential human decision, from how farmers plan their growing seasons to how governments issue public health warnings to how companies manage their supply chains. Despite this, most data science work treats weather as a given input rather than something worth understanding from the ground up. This project takes the opposite approach, working directly from raw daily observations across 257 cities to build, interrogate, and forecast from the data itself.

This report presents a full-stack analysis of the Global Weather Repository - a dataset of 137,998 daily city-level observations spanning 257 cities, 211 countries, and two years of coverage from May 2024 to April 2026. The analysis proceeds in five substantive stages: data quality diagnostics, preprocessing and feature engineering, exploratory data analysis, time series investigation, and predictive modelling.

The primary forecasting target is global daily mean temperature - the aggregate of all city-level readings on each date. This choice is deliberate because rather than building 257 separate city-level models, aggregation to the global mean leverages the full dataset, smooths city-specific noise, and produces a single coherent forecasting problem on which time series and machine learning approaches can be compared on equal terms.

2. Data Description

2.1 Dataset Overview

The raw dataset contains 41 columns across 137,998 rows. Before any preprocessing, a full diagnostic pass established the structure of the data. Column types break down into 30 numeric and 11 categorical columns. The categorical columns include operationally important fields - `condition_text` (49 unique values), `moon_phase` (8 unique values), `wind_direction` (16 unique values) - as well as the four astronomical time columns (`sunrise`, `sunset`, `moonrise`, `moonset`) which are stored as bare time strings with no date component, a structural issue addressed in preprocessing. The column type breakdown is documented in Appendix Figure 1.

Observations per city range from 1 to 710, with a median of 707 and a mean of 537. The wide standard deviation (285) reflects an uneven collection schedule: a majority of cities have near-complete two-year coverage while a tail of sparsely represented cities brings the mean down considerably. This heterogeneity in coverage depth has direct implications for the anomaly detection and time series modelling steps, where minimum observation thresholds must be enforced.

2.2 Data Quality Diagnostics

No true NaN values were present in the raw data and this is something worth noting because it complicates rather than simplifies the preprocessing task. The absence of explicit nulls means that data quality issues, where they exist, are disguised as valid-looking values rather than flagged as missing. Two categories of corruption were identified.

First, physically impossible values. A systematic range check grounded in actual physical bounds flagged 11 values across 8 columns for replacement with NaN (Appendix Figure 2). The rationale for using physical rather than statistical bounds is important: using dataset min/max as validity thresholds would mean accepting the corrupt values that happen to be the most extreme observations, since they define the bounds. For air quality columns, the lower bound of zero is both physically correct (concentrations cannot be negative) and simultaneously catches the -9999 sentinel values present in Carbon Monoxide and Sulphur Dioxide without requiring a hardcoded sentinel list.

Second, one logical duplicate was identified: a row sharing the same city, country, and timestamp as another which suggests a repeated API pull rather than a real second observation. It was retained rather than dropped, as the information content of such a duplicate is negligible rather than distorting.

- Full-row duplicates: 0.
- Logical duplicates (same city and timestamp): 1.
- Physically impossible values replaced with NaN: 11.
- Remaining NaN after imputation: 0 (discussed in section 3.1).

2.3 Country Name Standardisation

A non-trivial but easily overlooked data quality issue emerged in the country column: 21 of the 211 unique country names were stored in non-English scripts, foreign-language spellings, or inconsistent formatting. Detection used a canonical English country list built directly from the dataset itself — any name not present on that list was flagged automatically, with no reliance on character filters or observation count heuristics. The flagged entries included Arabic-script entries (كولومبيا for Colombia), Cyrillic entries (Гватемала for Guatemala, Польша and Полônia for Poland, Турция for Turkey), French spellings (Estonie for Estonia, Inde for India, Mexique for Mexico), German spellings (Jemen for Yemen, Komoren for Comoros, Saudi Arabien for Saudi Arabia), Spanish and Portuguese spellings (Letonia for Latvia, Marrocos for Morocco), a transliteration variant (Kyrgyzstan for Kyrgyzstan), and a formatting inconsistency (USA United States of America as a duplicate of the existing United States of America entry). Each was verified manually against a canonical list before correction — automated fuzzy string matching was considered but rejected because cross-language translation cannot be resolved by

string similarity alone. The mapping reduced the effective country count from 211 to 191 after consolidating duplicates (Appendix Figure 3).

Country representation is highly skewed as Bulgaria leads with 1,636 observations while Libya, Kosovo, and Togo each contribute a single entry (Appendix Figure 4). Colombia's count of only 2 observations is itself a data quality signal: the Arabic-script entry كولومبيا was consolidated into Colombia during standardisation, meaning what appeared to be two separate countries was in fact one split across two spellings. Countries with very few observations should be excluded from any country-level analysis, as a single reading cannot support any meaningful inference.

3. Methodology

3.1 Preprocessing

Preprocessing follows a deliberate order, where each step is designed to produce a clean foundation for the next rather than operating independently. The sequence is: drop redundant columns, replace physically impossible values, detect anomalies, then impute remaining NaN.

Eight columns were dropped - the full set of imperial unit equivalents and `last_updated_epoch`. Before dropping, the conversion formulas were verified empirically to confirm these columns carried no independent information beyond their SI counterparts (Appendix Figure 5). Keeping both would cause perfect multicollinearity in linear models and dilute feature importance in tree-based models.

Anomaly detection compares each observation to the same city's reading exactly one year prior rather than to the global distribution because if we take the example of a 45°C day in Doha it should then be compared to other Doha summers, not to naturally cooler cities such as in Norway. Any observation more than 4 standard deviations from that year-ago baseline is flagged. This threshold is conservative: a genuine 4σ deviation occurs by chance roughly once in 15,000 observations. The result was 1,256 anomalies flagged (0.82%), the vast majority involving only one variable behaving abnormally, confirming these are isolated sensor errors rather than wholesale station failures (Appendix Figure 6).

Imputation used forward-fill within each city, so each missing value takes the most recent prior reading from the same location. This preserves temporal continuity in a way that global mean imputation cannot: a city's missing January reading should inherit from its own December reading, not from the mean of 256 other cities with completely different climates.

3.2 Feature Engineering

Feature engineering was guided by the principle that every new variable must encode information the raw columns do not already provide in a model-accessible form. Features fall into five categories.

Cyclical encoding addresses the circular nature of temporal and directional variables. Hour 23 and hour 0 are adjacent in physical reality but 23 units apart if encoded as integers. Projecting hour, day-of-year, month, and wind direction onto the unit circle via sin/cos pairs preserves the correct neighbourhood structure for any distance-based or gradient-based model. This applies equally to wind direction: 359° and 1° are nearly the same compass heading, but an integer encoding treats them as maximally distant.

Astronomical features such as daylight hours and solar noon deviation encode latitude and season simultaneously in a single number. Two cities at vastly different latitudes can have similar temperatures on a given date while having fundamentally different solar energy budgets. Daylight hours captures this joint effect in a way that month alone cannot. Solar noon deviation, computed from the midpoint between each city's own sunrise and sunset rather than a fixed 12:00, captures the diurnal temperature curve precisely regardless of longitude.

Log and square-root transforms address severe right skewness in air quality and precipitation columns. PM2.5 skewness of 9.1, PM10 at 16.8, Carbon Monoxide at 13.3, and precipitation at 17.7 all received log1p transformation. Ozone at 1.19 was an exception: log1p over-corrected it while square-root brought it to -0.31 - a much milder compression that avoids the over-correction problem.

Lag features and rolling statistics give models access to recent trajectory rather than only current state. A pressure trend feature (today minus yesterday) is meteorologically motivated: falling pressure is a classical storm precursor regardless of the absolute pressure level. PM2.5 momentum captures whether air quality is improving or deteriorating. Rolling 7-day mean and standard deviation of temperature, humidity, and pressure smooth short-term noise while encoding whether conditions have been stable or volatile recently.

Composite indices such as the Human Comfort Index and the WHO-weighted Air Quality Health Risk Score are original features that do not exist in the raw dataset. The Comfort Index combines temperature, humidity, wind, UV, and air quality into a single normalised score with weights reflecting relative physiological impact. The AQ Health Risk Score uses WHO evidence weights to combine six pollutant readings into a continuous composite, giving models finer resolution than the coarse 1–6 EPA/DEFRA ordinal scales already in the data.

3.3 Multicollinearity Check - VIF Analysis

Before finalising the linear regression feature set, Variance Inflation Factor (VIF) analysis was applied iteratively. Features with $VIF > 10$ were removed one at a time, recomputing VIF after each removal, until all remaining features satisfied the threshold. Fourteen features were removed in this process, with the highest initial VIFs reflecting expected relationships: `pressure_mb` (893.1) was removed first as it was highly correlated with daylight-hours, dew point, and temperature through their shared seasonal structure, followed by `air_quality_PM2.5_log` (147.4), `comfort_index` (138.5 - itself a composite of several other features), and `air_quality_Carbon_Monoxide_log` (101.1). The final linear regression feature set contains 17 features, all with VIF below 10 (Appendix Figure 7). The ML feature set adds 13 lag and rolling features on top of this, bringing the total to 30.

4. Exploratory Data Analysis

4.1 Global Temperature & Precipitation Trends

The global daily mean temperature follows a clear annual cycle, rising in summer and falling in winter, which is exactly what you would expect (Appendix Figure 8). The fact that this pattern appears cleanly after averaging across 257 cities in different countries and climates confirms the dataset is internally consistent. The 30-day rolling average is overlaid to smooth out day-to-day fluctuations and make the underlying seasonal shape easier to see.

Several sharp downward spikes are visible in the raw daily series - days where the global mean drops by several degrees in a single step before immediately recovering. These are probably data quality artefacts: on those days, one or a handful of cities reported corrupt readings far below any physically plausible temperature, pulling the global mean down. These correspond directly to the anomaly days identified in Section 4.5 and visible in the anomaly scatter plot (Appendix Figure 6). The fact that no forecasting model predicts these spikes is not a model failure, it is actually the correct behaviour. Unpredictable station-level data corruption we do not consider as a weather signal.

Precipitation at the global mean level is low and stable throughout the year, fluctuating between 0.05mm and 0.40mm daily average across the full city panel. This reflects the composition of the sample: many cities in the dataset are in arid and semi-arid regions where precipitation is rare, pulling the global mean consistently downward regardless of season.

4.2 Climate Zone Analysis

The simplified Koppen classification produces five well-separated temperature distributions that are meteorologically sensible and internally consistent (Appendix Figure 9). Tropical cities

cluster tightly above 25°C with minimal seasonal variation - consistent with their proximity to the equator and the small solar angle changes they experience across the year. Arid cities show the highest variance, reflecting the large diurnal swings characteristic of desert climates where low humidity allows rapid radiative cooling at night. Temperate and Continental zones overlap substantially in the centre of the distribution, which is expected given our 5-class system collapses distinctions the full 30-subcategory Koppen scheme would preserve - London and Melbourne, for instance, are both classified as Temperate despite having opposite seasonal cycles. Polar cities contribute only 41 observations and are excluded from any zone-level interpretation.

4.3 Air Quality & Environmental Drivers

The atmospheric inversion hypothesis which states that high pressure and low wind create conditions that trap pollutants near the surface is directly testable in this dataset. Hexbin plots show PM2.5 concentrations are highest at low wind speeds (below 20 km/h) and drop sharply as wind increases, consistent with dispersal (Appendix Figure 10). The correlation heatmap reinforces this: AQ health risk and wind speed carry a negative relationship, while falling pressure precedes precipitation as expected.

4.4 Categorical Variable Profiles

Condition text is dominated by Partly Cloudy (29.05%) and Sunny (28.40%), together covering over half of all observations which is consistent with a globally sampled dataset weighted toward arid and temperate cities (Appendix Figure 11).

Moon phase is approximately uniformly distributed. Waxing Gibbous days show the highest mean precipitation on non-zero days (0.430mm), but the differences across phases are small and uneven sample sizes make any causal inference about lunar effects on precipitation premature (Appendix Figure 12).

Wind tends to blow from the east most often as E, ENE, and ESE together make up 25.3% of all readings which makes sense given how many cities in the dataset are in tropical regions where easterly trade winds are the norm.

4.5 Anomaly Detection

Rather than flagging anomalies on raw values, which would incorrectly flag a hot summer day in Doha as unusual, we subtract each observation's value from the same city's reading exactly one year prior. What remains is the deviation from the seasonal baseline. Any observation more than 4 standard deviations from that baseline is flagged as anomalous.

This approach identified 1,256 observations (0.82%) across 298 days (Appendix Figure 6). They cluster slightly in late 2025 and early 2026, though whether this reflects genuine station deterioration or a statistical artefact is unclear. Importantly, 1,242 of the flagged observations have only one variable reading abnormally - suggesting an isolated sensor error rather than a full station failure. The rest of the row remains usable; only the flagged column's value should be treated with caution.

Almost no anomalies are flagged in the first year of the dataset, while the second year is heavily flagged. This is a direct consequence of the methodology: the first year serves as the reference baseline and the second year is compared against it.

4.6 Spatial Patterns

The geographic extremes confirm expected findings: Reykjavik is the northernmost city, Wellington the southernmost (Appendix Figure 13). Notably, the easternmost city (Funafuti, Tuvalu at 179.22°E) and westernmost (Nuku'alofa, Tonga at -175.20°W) are Pacific neighbours despite sitting at opposite ends of the longitude scale, precisely why longitude requires cyclical encoding.

The comfort map uses five colour bands from green (comfortable) to red (very uncomfortable). The thresholds were tuned to the actual distribution of scores (which range from 0.33 to 0.90 with most cities clustering around 0.75) because using round-number cutoffs produced almost no variation across cities. With calibrated thresholds, Kuwait (0.33) and Ar Riyadh (0.43) stand out as the least comfortable cities globally, while European and Oceanian cities dominate the top.

Before fitting any time series model, two diagnostic steps were performed to directly inform the SARIMA specification: a stationarity test and an autocorrelation analysis.

The Augmented Dickey-Fuller (ADF) test applied to the raw global daily mean temperature series yields a p-value of 0.385, confirming the series is non-stationary (Appendix Figure 14). After taking one difference, the p-value drops to 0.0001, confirming stationarity. This establishes $d = 1$ as the correct differencing order. Taking a second difference would be unnecessary and would artificially introduce negative autocorrelation, degrading model performance.

The ACF and PACF plots on the differenced series show a sharp cutoff in the PACF after lag 1, consistent with AR(1), and a pattern in the ACF consistent with MA(1), confirming the SARIMA(1,1,1) non-seasonal specification (Appendix Figure 15). A weekly seasonal period ($s = 7$) is used rather than annual ($s = 365$) because with only 560 training observations, fitting an annual seasonal component would require estimating more parameters than the data can support.

6. Model Building & Forecasting

6.1 Model Architecture

Six models are built across two parallel tracks. Time series models: SARIMA and Holt-Winters operate on the global daily mean temperature series of approximately 560 training and 151 test days. Supervised machine learning models: Linear Regression, Random Forest, XGBoost, and a feedforward Artificial Neural Network are trained on city-level observations (122,068 training rows and 30,483 test rows), with predictions aggregated to the global daily mean for comparison.

Training ML models at city level rather than on the 560-point global series is deliberate because the global series alone is too small to train a neural network effectively, while city-level data provides 122,068 observations for the model to learn from. The aggregation step also smooths individual city errors, which is why global-level MAEs for ML models are very low. The train/test split is strictly temporal: the first 80% of the timeline is used for training and the final 20% (December 2025 through April 2026) for testing. Shuffling observations before splitting would introduce look-ahead bias by allowing the model to train on future observations and test on past ones which is a form of data leakage that would produce misleadingly optimistic evaluation metrics.

6.2 Results

The full model comparison is presented in Appendix Figure 17. The results reveal a stark two-tier performance structure.

Model	MAE (°C)	RMSE	MAPE (%)	R ²	MASE	DirAcc (%)
Ensemble (selective)	0.024	0.037	0.18	1.000	0.033	98.0
Random Forest	0.025	0.049	0.21	0.999	0.034	96.0
ANN	0.037	0.043	0.23	1.000	0.051	98.0
XGBoost	0.061	0.090	0.42	0.998	0.083	94.7
Linear Regression	1.209	1.577	8.79	0.389	1.646	74.0
SARIMA (1,1,1)(1,0,1,7)	1.573	2.076	11.30	-0.059	2.141	43.3
Holt-Winters (s=30)	2.659	3.459	16.28	-1.940	3.620	43.3

Table 1: Model performance on the global daily mean temperature test set (December 2025 - April 2026). MAE in degrees Celsius.

The results split clearly into two tiers. ML models achieve MAEs below 0.065°C , while SARIMA and Holt-Winters exceed 1.5°C . This is not so much about model quality but more so about data availability: the time series models are working with a fraction of the information available to the ML models. Both time series models score MASE above 2.0, meaning they perform worse than simply predicting today's temperature as yesterday's. Their directional accuracy of 43.3% is below the 50% coin-flip threshold, suggesting they are not just imprecise but also systematically wrong about the direction of day-to-day change.

Random Forest is the best individual model (MAE = 0.025°C , directional accuracy = 96%), narrowly ahead of ANN and XGBoost. The selective ensemble combines all three ML models using inverse-MAE weighting: models that performed better receive a higher weight, and the ensemble achieves the best results overall (MAE = 0.024°C , directional accuracy = 98%). The logic behind the ensemble is that different models make different errors: Random Forest, ANN, and XGBoost each use different learning mechanisms, so their mistakes do not perfectly overlap. Averaging them causes those errors to partially cancel out, producing a more stable and accurate forecast than any single model alone. SARIMA and Holt-Winters are automatically excluded from the ensemble by the $3\times$ MAE threshold, since including poorly performing models would add noise rather than reduce error.

The most underappreciated result is Linear Regression's directional accuracy of 74% - well above naive and the time series models - despite using no lag features and only 17 variables. This suggests the engineered meteorological features carry genuine predictive signal for temperature direction even within a purely linear framework.

6.3 Feature Importance

Three importance methods were applied to assess which features drive predictions (Appendix Figures 18, 19, 20).

Permutation importance identifies yesterday's temperature (`temperature_celsius_lag1`) as by far the strongest predictor confirming that temperature is highly persistent day-to-day. The engineered features `dew_point` and `heat_humidity_stress` also rank highly, validating that the composite meteorological features capture information beyond what the raw variables provide individually.

SHAP values confirm this ranking while adding directional insight. Yesterday's temperature pushes today's prediction higher when it was warm and lower when it was cold - essentially a

one-to-one relationship that reflects how persistent temperature is day-to-day. Dew_point behaves differently: its impact on predictions changes depending on what other conditions are present that day, rather than having a consistent fixed effect.

The linear regression coefficients reveal a less intuitive finding: aq_health_risk carries the largest coefficient by far. Air quality does not directly cause temperature: the explanation is that after VIF analysis removed several collinear predictors like pressure_mb and daylight_hours, aq_health_risk absorbed their explanatory role as a correlated proxy.

7. Interpretation & Conclusions

The most important finding in this analysis is not the MAE numbers, it is actually what they reveal about the nature of the forecasting problem itself. The global daily mean temperature is a smooth, heavily averaged series that is genuinely difficult to improve upon with a classical time series model, not because SARIMA is poorly specified but when 257 cities are collapsed into a single daily number, the result is so persistent that yesterday's value is already a strong predictor of today's. Both time series models fail to beat that naive baseline. The ML models succeed largely because they sidestep this problem entirely as it trains on the raw city-level observations rather than the processed global mean, and then averaging predictions at the end.

The feature importance results tell us that yesterday's temperature dominates all three importance methods, which confirms temperature persistence rather than revealing anything new. The more interesting finding is that engineered features such as dew_point, heat_humidity_stress, and feels_delta consistently outrank most of the original 41 columns. The atmosphere's thermodynamic state, captured through interaction between heat and humidity for example, carries predictive information that neither temperature nor humidity encodes individually. This validates the feature engineering effort and suggests that for weather forecasting, composite physical indices are genuinely informative inputs.

Beyond the modelling results, several findings emerged as well. The comfort index produced some striking geographic findings. Kuwait ranked as the least comfortable city globally, driven by extreme heat combined with poor air quality. Ar Riyadh followed closely. At the other end, European and Oceanian cities dominated the top of the comfort rankings, with temperate climates and cleaner air giving them a consistent advantage year-round. The spatial maps also revealed that the cities with the highest air quality health risk are concentrated in South and East Asia, particularly in cities with high industrial activity and dense urban populations.

The continent radar chart (Appendix Figure 16) shows Africa scoring highest on temperature and lowest on comfort, while Europe presents the most balanced profile across all six dimensions. Asia shows the most asymmetric shape: moderate on temperature but by far the worst on PM2.5 thus reflecting the concentration of high-pollution industrial cities in the sample.

The moon phase analysis warrants one final note. A pattern exists in the data: Waxing Gibbous days show higher mean precipitation than Full Moon days but the sample sizes across phases are too uneven to draw conclusions. This is worth stating plainly rather than either overstating a finding or dismissing it entirely. The dataset suggests a pattern but cannot confirm it.

8. Limitations

- The ML models report near-perfect results ($MAE = 0.024^{\circ}C$, $R^2 = 1.000$), but this might be partly because errors across 257 cities cancel each other out when averaged. A city predicted $2^{\circ}C$ too high and another $2^{\circ}C$ too low together produce a perfect global mean prediction. City-level accuracy would tell a more complete story.
- The ML models rely heavily on yesterday's temperature to make predictions, which is already a strong predictor given the global mean moves by only $0.3^{\circ}C$ per day on average. City-level evaluation would be much harder to do but would offer more realistic results. Although currently this produces accurate results it raises the question of how much the other features are genuinely contributing beyond simple temperature persistence.
- SARIMA uses a weekly seasonal period ($s = 7$) not because global temperature follows a seven-day cycle, but because the sample size makes an annual period infeasible. This is a practical workaround rather than a meteorologically motivated choice.
- Country standardisation was resolved through manual verification against a canonical list. This worked for the current dataset but would need to be repeated if the data is updated, since there is no fully automated solution that reliably translates country names across languages.
- Two years of data is too short for any meaningful climate analysis. All trend observations in this report describe this specific observation window only and should not be generalised.

9. Appendix

Figure 1: Column type summary

Shape: 137,998 rows x 41 columns

```
Column dtypes:
country                object
location_name          object
latitude               float64
longitude               float64
timezone               object
last_updated_epoch     int64
last_updated           object
temperature_celsius    float64
temperature_fahrenheit float64
condition_text         object
wind_mph               float64
wind_kph               float64
wind_degree            int64
wind_direction         object
pressure_mb            float64
pressure_in            float64
precip_mm              float64
precip_in              float64
humidity               int64
cloud                  int64
feels_like_celsius     float64
feels_like_fahrenheit float64
visibility_km          float64
visibility_miles       float64
uv_index               float64
gust_mph               float64
gust_kph               float64
air_quality_Carbon_Monoxide float64
air_quality_Ozone      float64
air_quality_Nitrogen_dioxide float64
air_quality_Sulphur_dioxide float64
air_quality_PM2.5      float64
air_quality_PM10       float64
air_quality_us-epa-index int64
air_quality_gb-defra-index int64
sunrise                object
sunset                 object
moonrise                object
moonset                 object
moon_phase             object
moon_illumination       int64
```

Figure 2: Physically impossible values replaced per column

```
temperature_celsius: 1 replaced
wind_kph: 1 replaced
gust_kph: 1 replaced
pressure_mb: 2 replaced
uv_index: 2 replaced
air_quality_Carbon_Monoxide: 1 replaced
air_quality_Sulphur_dioxide: 1 replaced
air_quality_PM10: 2 replaced
```

Total values replaced with NaN: 11

Figure 3: Non-standard country names detected

Non-standard country names detected (21):

```
'Bélgica'
'Estonie'
'Inde'
'Jemen'
'Komoren'
'Kyrgyzstan'
'Letonia'
'Malásia'
'Marrocos'
'Mexique'
'Polônia'
'Saint-Vincent-et-les-Grenadines'
'Saudi Arabien'
'Südkorea'
'Turkménistan'
'USA United States of America'
'Гватемала'
'Польша'
'Турция'
'كولومبيا'
'火鸡'
```

Figure 4: Country representation - most and least represented countries with bar charts

Most represented countries (most entries):

country	
Bulgaria	1636
Indonesia	1419
Thailand	1418
Turkey	1416
Sudan	1414
Bolivia	1407
Iran	1374
Belgium	1329
Madagascar	1167
Vietnam	1162

Least represented countries (fewest entries):

country	
Guatemala	703
Nicaragua	703
Costa Rica	702
El Salvador	702
Cote d'Ivoire	248
Colombia	2
Lao People's Democratic Republic	1
Libya	1
Kosovo	1
Togo	1

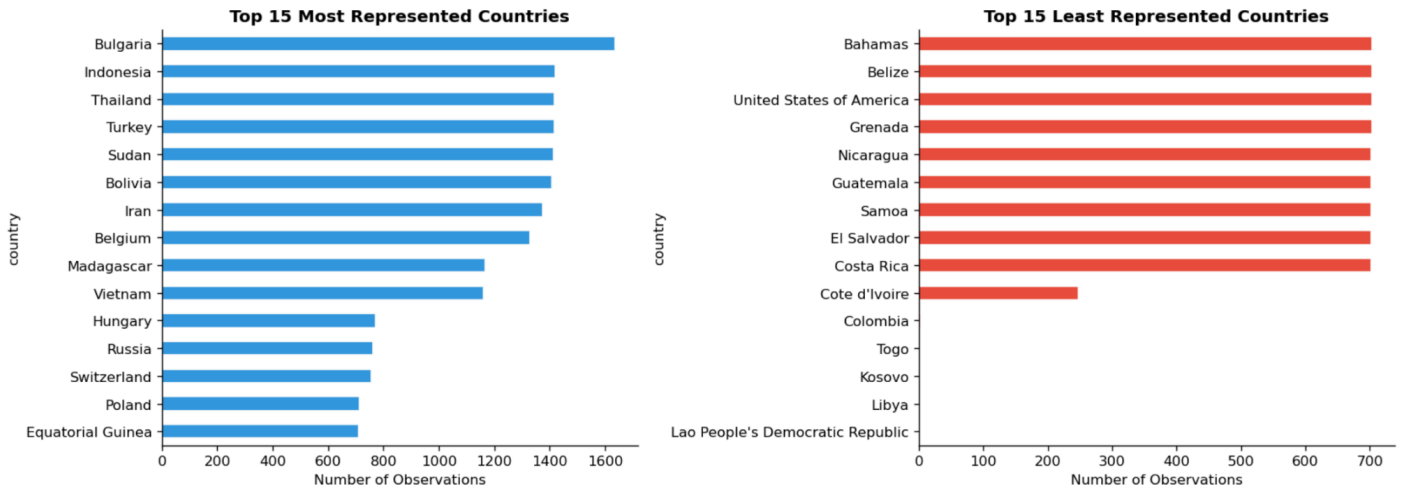


Figure 5: Temperature conversion verification

Temperature C→F conversion residuals:

Mean residual : 0.033102°F

Max residual : 0.120000°F

Rows > 0.5°F : 0

Conclusion: Conversion is consistent across all rows.

Fahrenheit columns are pure redundant transforms – safe to drop.

Figure 6: Anomaly detection

Anomalies flagged: 1,256 (0.82%)

Breakdown by number of simultaneously anomalous columns:

anomaly_col_count

0 151295

1 1242

2 14

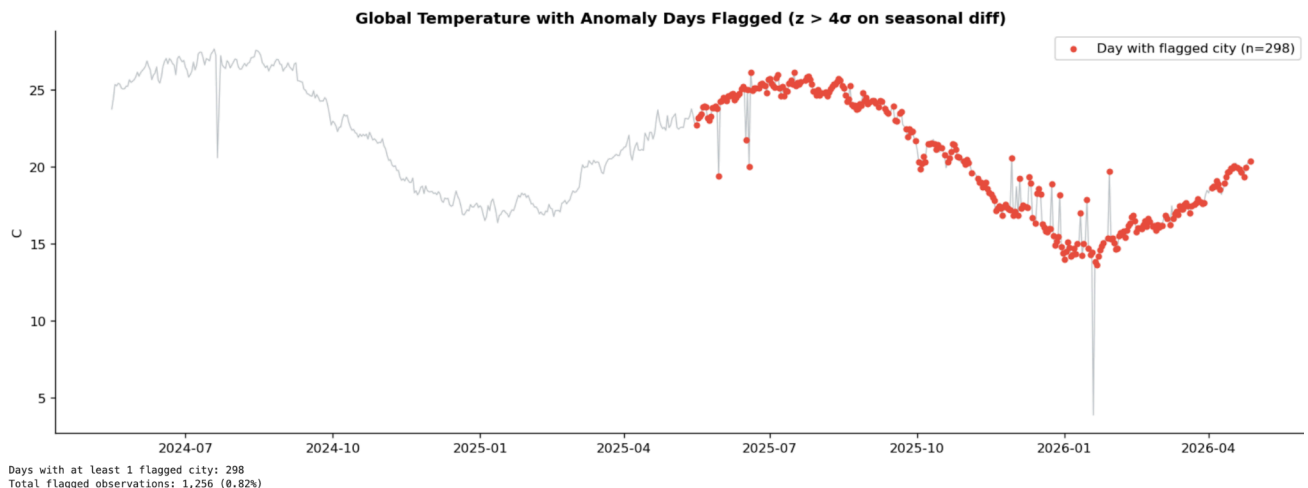


Figure 7: VIF analysis

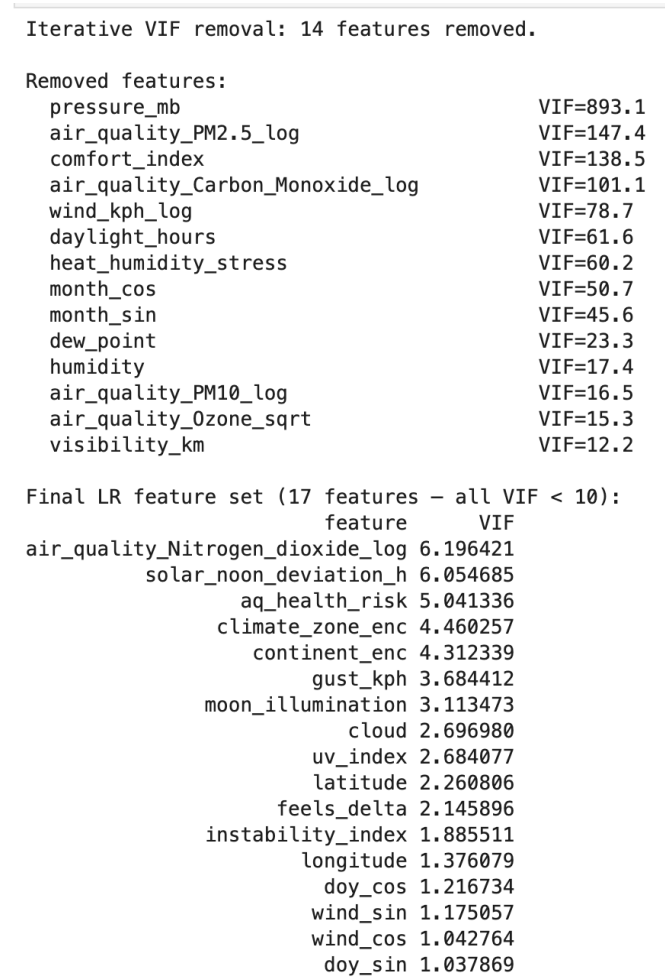


Figure 8: Global mean temperature and precipitation trends, 2024-2026

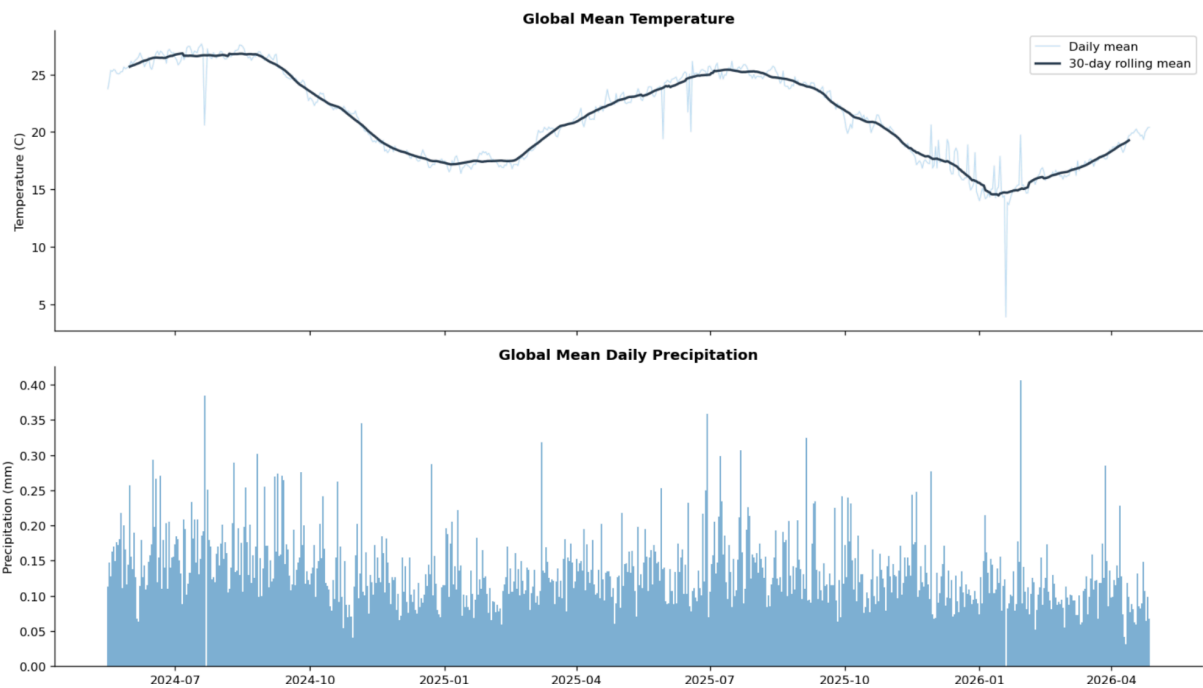


Figure 9: Temperature distribution by Koppen climate zone + log1p/sqrt transform validation

Climate zone distribution:		Column	Raw	Transformed
climate_zone		air_quality_PM2.5	8.65	0.18 (log1p)
Temperate	66996	air_quality_PM10	17.42	0.57 (log1p)
Tropical	43238	air_quality_Carbon_Monoxide	13.05	1.06 (log1p)
Arid	32905	air_quality_Nitrogen_dioxide	3.17	0.36 (log1p)
Continental	9371	precip_mm	17.70	4.13 (log1p)
Polar	41	wind_kph	1.54	0.00 (log1p)
		air_quality_Ozone	1.19	-0.31 (sqrt)

Figure 10: Meteorological drivers of air quality - hexbin plots

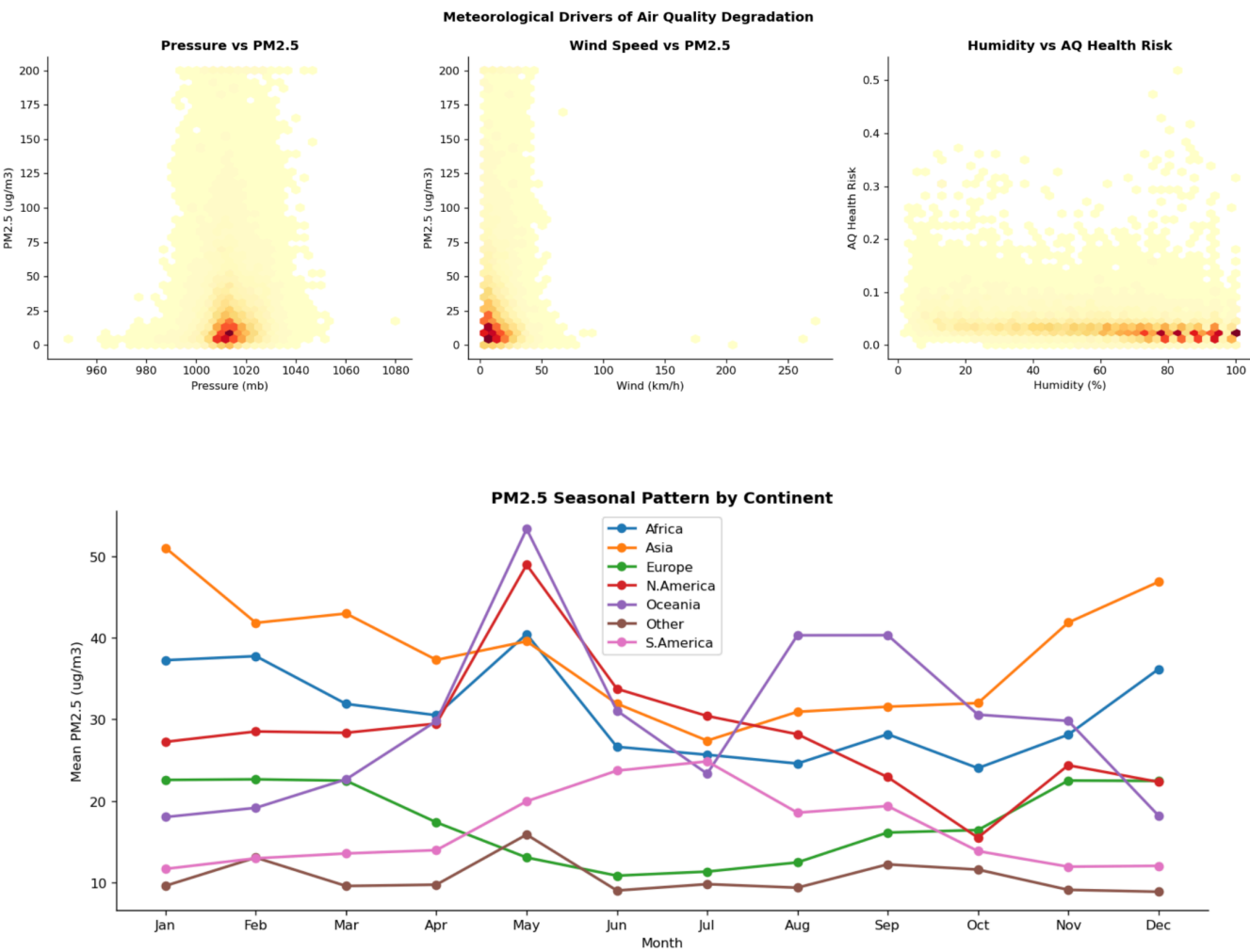


Figure 11: Categorical frequency analysis

— condition_text (49 unique values) —		
	count	%
condition_text		
Partly cloudy	44310	29.05
Sunny	43324	28.40
Patchy rain nearby	12884	8.45
Partly Cloudy	9705	6.36
Overcast	8561	5.61
Clear	8111	5.32
Mist	6106	4.00
Light rain	4816	3.16
Light rain shower	3468	2.27
Fog	1917	1.26
— moon_phase (8 unique values) —		
	count	%
moon_phase		
Waning Gibbous	40447	26.51
Waxing Gibbous	30884	20.25
Waning Crescent	30831	20.21
Waxing Crescent	30345	19.89
Full Moon	5122	3.36
Last Quarter	5046	3.31
New Moon	5002	3.28
First Quarter	4874	3.19
— wind_direction (16 unique values) —		
	count	%
wind_direction		
E	14621	9.58
ENE	12163	7.97
ESE	11764	7.71
NNE	10176	6.67
N	10150	6.65
S	9416	6.17
SSE	9209	6.04
NNW	9148	6.00
WSW	8732	5.72
SW	8609	5.64

Figure 12: Moon phase × mean precipitation

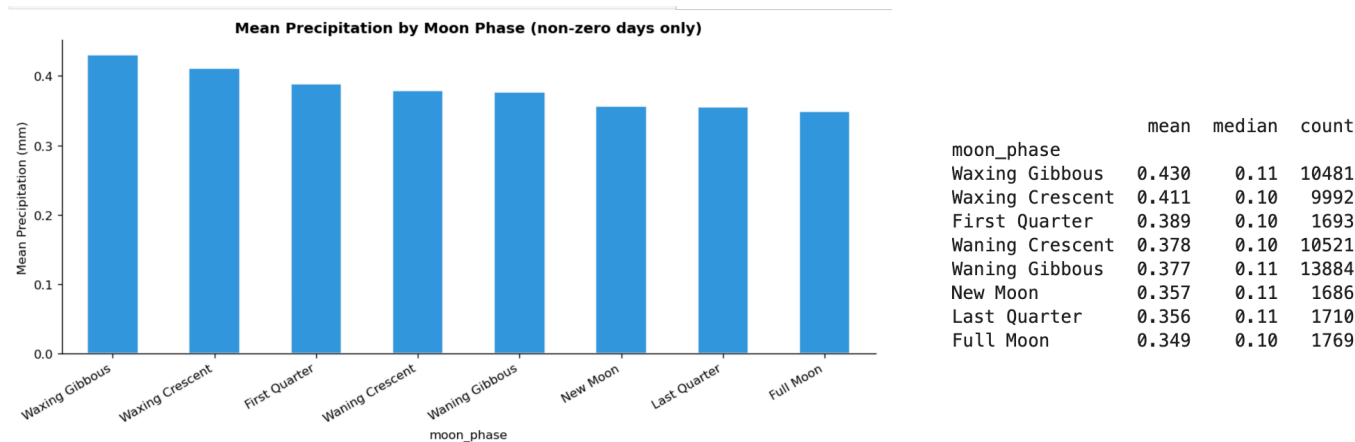


Figure 13: Geographic extremes - northernmost, southernmost, easternmost, westernmost cities with coordinates

Northernmost cities (highest latitude):

location_name	country	lat
Reykjavik	Iceland	64.150000
Grindavik	Iceland	63.830000
Vestmannaeyjar	Iceland	63.366203
Helsinki	Finland	60.176392
Oslo	Norway	59.917291

Southernmost cities (lowest latitude):

location_name	country	lat
Wellington	New Zealand	-41.300000
Melbourne	Australia	-37.820000
Canberra	Australia	-35.282792
Montevideo	Uruguay	-34.858443
Buenos Aires	Argentina	-34.587948

Easternmost cities (highest longitude):

location_name	country	lon
Funafuti	Tuvalu	179.217294
Suva	Fiji Islands	178.417293
Wellington	New Zealand	174.782901
Majuro	Marshall Islands	171.382706
Tarawa	Kiribati	169.532714

Westernmost cities (lowest longitude):

location_name	country	lon
Nuku'alofa	Tonga	-175.200000
Nuku'Aloia	Tonga	-175.200000
Apia	Samoa	-171.732923
Washington Harbor	United States of America	-123.043216
Washington Park	United States of America	-120.490000

Figure 14: ADF stationarity test

Raw global daily mean temperature

ADF statistic : -1.7900
p-value : 0.3854
Critical (5%) : -2.8657
Conclusion : NON-STATIONARY – fail to reject H0

First difference (d=1)

ADF statistic : -4.6112
p-value : 0.0001
Critical (5%) : -2.8657
Conclusion : STATIONARY – reject H0

Figure 15: ACF and PACF plots

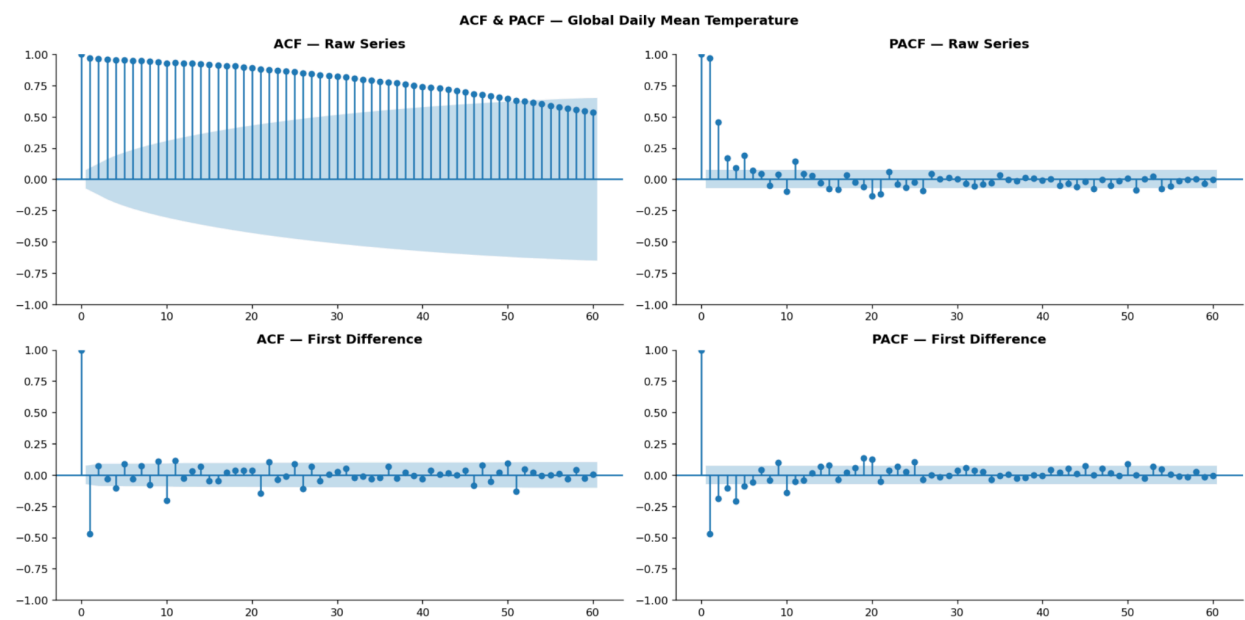


Figure 16: Continent radar chart and table

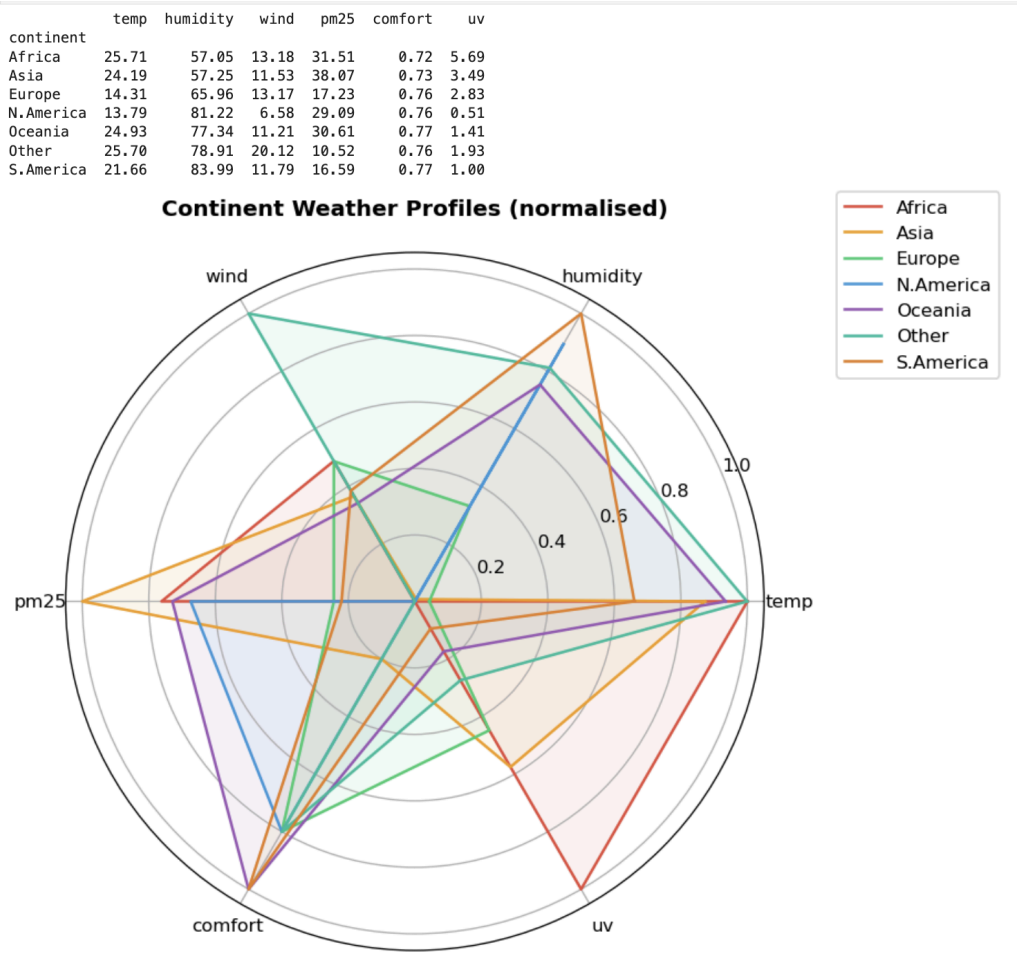


Figure 17: Model comparison table and forecast residuals

Final Model Comparison:

	MAE	RMSE	MAPE	R2	MASE	DirAcc
model						
Ensemble (selective)	0.024	0.037	0.177	1.000	0.033	98.000
Random Forest	0.025	0.049	0.211	0.999	0.034	96.000
ANN	0.037	0.043	0.228	1.000	0.051	98.000
XGBoost	0.061	0.090	0.423	0.998	0.083	94.667
Linear Regression	1.209	1.577	8.794	0.389	1.646	74.000
SARIMA(1,1,1)(1,0,1,7)	1.573	2.076	11.299	-0.059	2.141	43.333
Holt-Winters (s=30)	2.659	3.459	16.280	-1.940	3.620	43.333

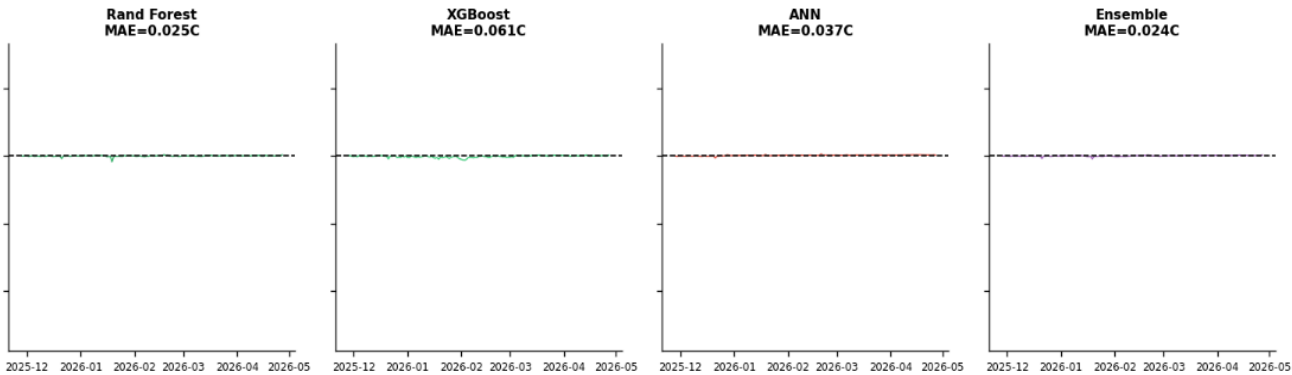
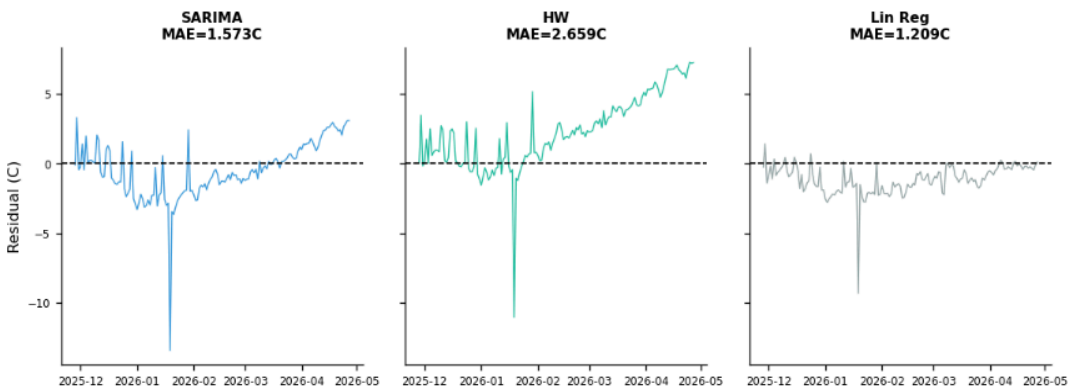


Figure 18: Permutation importance - XGBoost, top 20 features with error bars

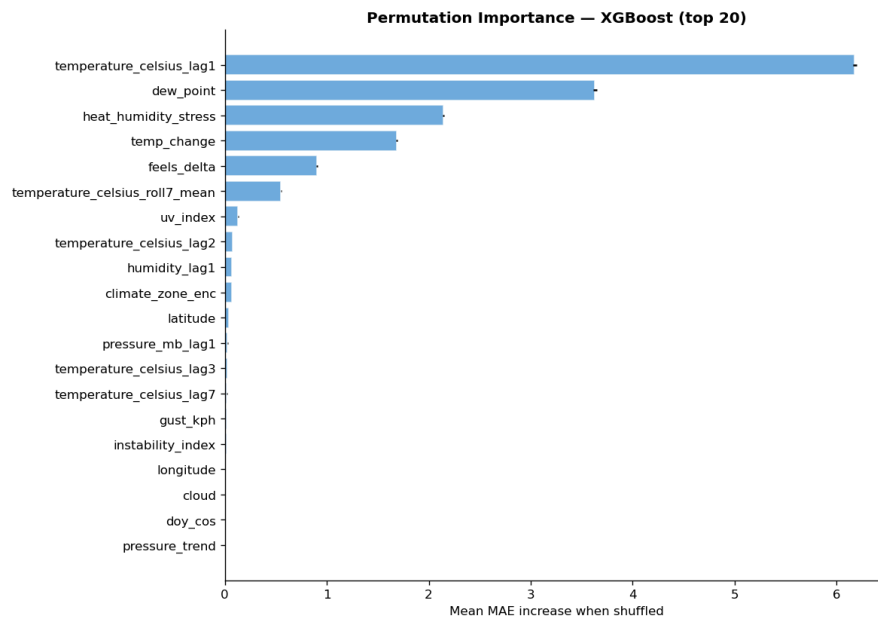


Figure 19: SHAP summary and dependence plot - XGBoost temperature forecast

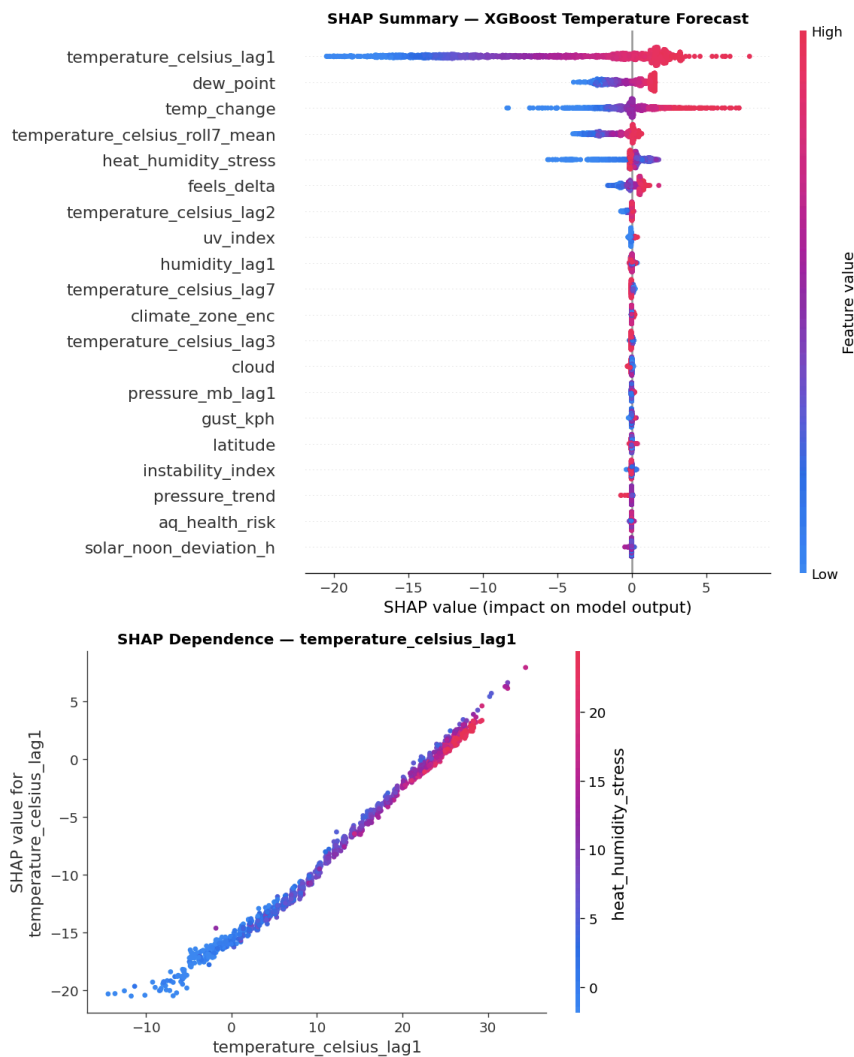


Figure 20: Linear regression coefficients - top 15 by magnitude

